Dong 2020 neuroblastoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/13 tumours (≥100 cells) intermix on one continuous manifold (1 kNN component; iLISI source 1.00→1.38, sample 3.38 of 13). The two capture groups mix through the body; the small near-pure pockets are the two largest MYCN-amplified tumours — genuine CNV states, not a per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

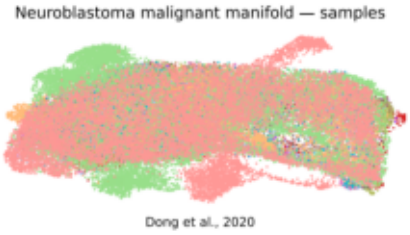
Single pSom→Crest axis: pSom high across the body, Crest anti-correlated (r=−0.65) as an inverse gradient (joint pSom&Crest-high enrichment 0.38); Gast marks the origin, declining along pseudotime (ρ=−0.63). No hub-and-spoke divergence — gradient, not hub.

Gate 3 — cross-repression → NA

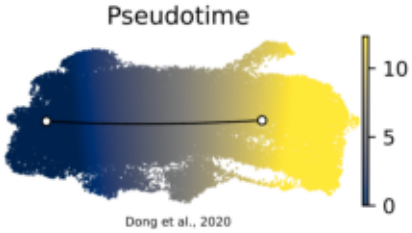
Non-branching: PPT forks scale with node count and the curve is 2-tip / 0-fork, so there is no bifurcation and no Mixed-type hub to test.

STEP FIGURES

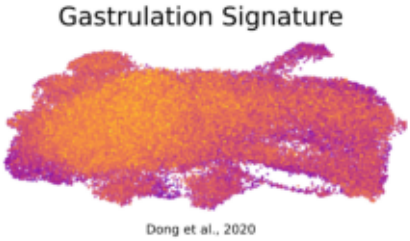
Manifold — samples



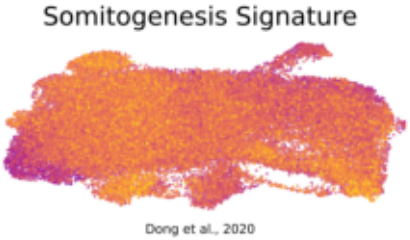
Pseudotime



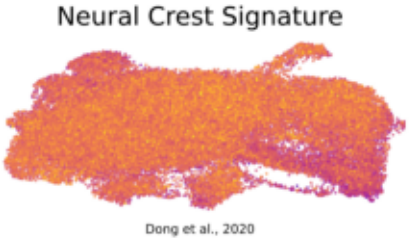
Gastrulation



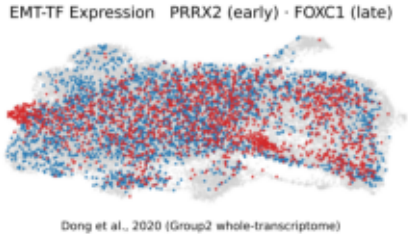
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Neuroblastoma (Dong et al., 2020), 10x — 13 tumours / 55,190 cells; malignant only: 47,384 cells (all 13 tumours ≥100) enter preprocessing. Captured in two panels; the joint manifold is built on the 12,113-gene shared panel so all tumours share one feature space.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The neuroblastoma malignant manifold is one continuous gastrulation/somitogenesis → neural-crest-inverse axis (low-cycling origin → proliferative terminus), matching developmental order; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub. The curve tracks proliferation (pt↔cycling ρ=+0.80), but the modules are cell-cycle-free (Gast 0/81, Crest 0/104 cc-genes) and pSom↔Crest anti-correlation survives in non-cycling cells (r=−0.60) — developmental, so the whole manifold is kept (cycling co-variation a caveat). EMT-TFs PRRX2 (early) → FOXC1 (late, ρ=+0.15).

Parameter choices

Min cells	≥100 floor (13/13; gap 1.06×)	**Integration**	Harmony θ=10; 20 PCs; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Compartments**	Body 35,196 / Cycling 12,188 (modules cc-free)
Geometry	Linear (non-branching); 2 tips / 0 forks	**EMT-TFs**	PRRX2 (early) → FOXC1 (late); Group2 whole-transcriptome